

高级人工智能

第一章 绪论

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Outline

- What is AI?
- A brief history
- The state of the art
- 课程介绍

教材

- S. Russell and P. Norvig *Artificial Intelligence: A Modern Approach* Prentice Hall, 2010, **Third Edition**, 早期版本也没问题,我们会补充些新的内容

What is AI?

Views of AI fall into four categories:

Thinking humanly

"The exciting new effort to make computers think ... machines with minds, in the full and literal sense." (Haugeland, 1985)

"[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning .. ." (Hellman, 1978) 文

Thinking rationally

"The study of mental faculties through the use of computational models."

(Charniak and McDermott, 1985)

"The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992) 理

Acting humanly

"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990)

"The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991) 工

Acting rationally

"Computational Intelligence is the study of the design of intelligent agents."

(Poole et al, 1998)

"AI ... is concerned with intelligent behavior in artifacts." (Nilsson, 1998) 用

Thinking humanly: cognitive modeling

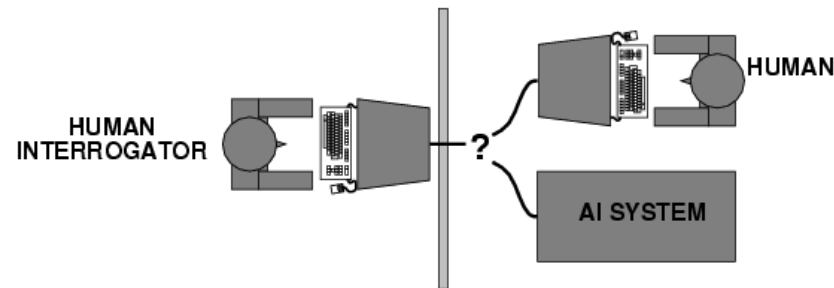
- 1960s "cognitive revolution": information-processing psychology
-
- Requires scientific theories of internal activities of the brain
- -- How to validate? Requires
 - 1) Predicting and testing behavior of human subjects (top-down)
 - or 2) Direct identification from neurological data (bottom-up)
- Both approaches (roughly, Cognitive Science and Cognitive Neuroscience) are now distinct from AI
-
- 相关研究：情感计算、评价挖掘、舆情分析...

Thinking rationally: "laws of thought"

- Aristotle: what are correct arguments/thought processes?
-
- Several Greek schools developed various forms of *logic: notation* and *rules of derivation* for thoughts; may or may not have proceeded to the idea of mechanization
-
- 现代AI中的逻辑除了包含一阶逻辑之类的古典逻辑以外，还有非单调逻辑、自然语言逻辑、模糊逻辑、弗协调逻辑、各种推理（常识推理、归纳推理、时间推理、因果推理、不确定推理等）方法等。
- Problems:
 - Not all intelligent behavior is mediated by logical deliberation
- 中国人不重视大逻辑（但是喜欢小聪明）

Acting humanly: Turing Test

- Turing (1950) "Computing machinery and intelligence":
- "Can machines think?" → "Can machines behave intelligently?"
- Operational test for intelligent behavior



- Predicted that by 2000, a machine might have a 30% chance of fooling a lay person for 5 minutes
- Anticipated all major arguments against AI in following 50 years
- Suggested major components of AI: knowledge, reasoning, language understanding, learning
- 也称作：狭义AI

Acting rationally: rational agent

- **Rational** behavior: doing the right thing
-
- The **right** thing: that which is expected to maximize goal achievement, given the available information
-
- Doesn't necessarily involve thinking – e.g., blinking reflex – but thinking should be in the service of rational action
-

Rational agents

- An **agent** is an entity that perceives and acts
- This course is about designing rational agents
- Abstractly, an agent is a function from percept histories to actions:

-

$$[f: \mathcal{P}^* \rightarrow \mathcal{A}]$$

- For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance
- Caveat: computational limitations make perfect rationality unachievable
 - design best **program** for given machine resources

AI到底是什么？

- AI就是建造 rational agents?
 - 书本观点强调agent
 - 理性，可能掩盖/发扬人性的善/恶
 - 理性会导致“囚徒困境”，从而得到并非是最好的结果
 - AI and ethics: project Maven
 - 你们完全可以有自己的想法

弱AI和强AI

- Weak AI, or Narrow AI: focuses on one narrow task which usually requires human intelligence.
- Strong AI, or Artificial General Intelligence (AGI): aims for a universal algorithm for learning and acting in any environment, which is still largely a “dream”. AGI means a machine with sentience, consciousness and mind

Outline

- What is AI?
- A brief history
- The state of the art

AI prehistory - Foundations of AI

- Philosophy Logic, methods of reasoning, mind as physical system foundations of learning, language, rationality
- Mathematics Formal representation and proof algorithms, computation, (un)decidability, (in)tractability, probability
- Economics utility, decision theory
- Neuroscience physical substrate for mental activity
- Psychology thinking and actions of humans and animals, experimental techniques
- Computer engineering algorithms, programming, fast computers
- Control theory design systems that maximize an objective function over time
- Linguistics knowledge representation, grammar

人工智能历史(1)

- 1.萌芽期 (1943-1955)
- 2.诞生 (1956)
- 3.early enthusiasm, great expectations (1952-1969)
- 4.A dose of reality (1966-1973)
- 5.知识就是力量 (1969-1979)
- 6.AI成为工业(1980至今)

人工智能历史(2)

- 7. 神经网络复兴（1986至今）
- 8. AI成为科学（1987至今）
- 9. 智能体的出现（1995至今）
- 10. 大数据的兴起（2001至今）
- 11. 深度学习的兴起（2006至今）

萌芽期（1943-1955）

- McCulloch & Pitts proposed a model of artificial neurons (1943)
 - Any computable function could be computed by some network of connected neurons
- Turing articulated a completed vision of AI in “Computing Machinery and Intelligence” (1950)
 - Turing test, machine learning, genetic algorithm, and reinforcement learning

AI的诞生（1956）

- 世界上第一次正式的AI会议在美国的Dartmouth College召开，为期2月
- 会上John McCarthy 正式提出“Artificial Intelligence”这一术语
- 著名参加者：J.McCarthy、C.Shannon、M.Minsky、N.Wiener、W.McCulloch、S.Papert
- 2006年有一次50周年纪念会议，并展望下个50年
 - <http://www.dartmouth.edu/~ai50/homepage.html>

early enthusiasm, great expectations (1952-1969)

- 1952-55: Samuel 's checkers program disproved the idea that computers can only do what they are told to do (via reinforcement learning)
- 1956: Newell & Simon: Logic Theorist
- 1958: Newell和Simon: GPS.
- 1958: McCarthy 发明著名的人工智能语言Lisp
- 1959: Frank Rosenblatt提出感知器模型
- 1959: Minsky和McCarthy正式成立MIT AI Lab。
- 1962: McCarthy和Minsky分裂, McCarthy到Stanford, 1963年创建Stanford AI Lab

early enthusiasm, great expectations (1952-1969)

- 1963: M. Ross Quillian开创语义网络 (Semantic Nets)
- 1965: MIT的Joseph Weizenbaum研制出ELIZA
 - 用英语进行交互
 - 回答任何问题
- 1969: SRI研制出机器人Shakey
 - 具有运动、感知和问题求解能力
- 1970: Stanford的Terry Winograd等研制出SHRDLU
 - 接受自然语言命令
 - 操作积木块
- 1970: Colmerauer研制出PROLOG语言的解释系统。不久，爱丁堡大学的Warren实现了编译系统

A dose of reality (1966-1973)

- 1958: Newell和Simon: 四个预测
 - 十年内，计算机将成为世界象棋冠军
 - 十年内，计算机将发现或证明有意义的数学定理
 - 十年内，计算机将能谱写优美的乐曲
 - 十年内，计算机将能实现大多数的心理学理论
- 但是，当时的AI系统有3个缺陷：
 - 没有领域知识，机器翻译闹笑话
 - 1966: ALPAC的负面报告造成 美国政府取消对机器翻译的资助
 - Microworld升级面临组合爆炸：算法复杂性理论
 - 简陋的表示无法解决复杂问题
 - 1969: Minsky 和 Papert的感知机报告造成美国政府取消对神经网络研究的资助。
- 1972: DARPA取消Stanford大学机器人研究(Shakey)的资助。
- 1973: James Lighthill爵士的负面报告使得英国政府取消对AI研究的资助
 - “人工智能研究是不成功的，不值得政府资助。”
- 1976: DARPA 取消对语音识别研究的资助

知识就是力量（1969-1979）

- 1965-83: Feigenbaum和Lederberg启动 DENDRAL工程
- 1972: Feigenbaum等的Mycin工程启动
- 1976: Doug Lenat的数学积分系统AM (Automated Mathematician)
- 1977: SRI启动 PROSPECTOR 工程帮助地质专家探测和解释矿物.1978年发现钼矿脉
- 1977: IJCAI会议上Edward Feigenbaum正式提出知识工程作为一门学科

AI工业(1980至今)

- 专家系统在工业界广泛使用
- 1980: 第一届美国AI协会会议（AAAI）在Stanford召开。
- 1981: 日本政府宣布日本五代机（first-generation computer）计划（即智能计算机）
- 1991: Guido van Rossum 发明了Python
- 1992: 日本政府宣布五代机计划失败。

神经网络复兴（1986至今）

- 1982: John Hopfield 掀起神经网络的研究
 - Reinvention of back-propagation algorithm (e.g. in 1969, 1974, 1985)

AI成为科学（1987至今）

- 科学的要素：
 - 坚实的理论基础，不是ad hoc解决方法
 - 理论假设+实验验证，且可重复
 - 面向实际问题而不是Toy problems
- 很多AI分支学科成熟
 - 语音识别中HMM的成功应用
 - 神经网络发展成一门学科
 - 概率论、统计学习理论的广泛使用

智能体的出现（1995至今）

- 1993: Shoham提出AOP, Agent-Oriented Programming
- 1995: Vapnik提出SVM
- 1997: IBM 深蓝II (Deep Blue)击败Garry Kasparov

大数据的兴起（2001至今）

- 早期的AI重视算法甚于数据，但是现在可能要反过来：很多算法在数据充分时性能接近
- 早在1995年Yarowsky就发现，在做词义消歧的时候，不需要标注数据，仅用一本词典和大量的未标注数据即可达到96%的准确率
- 2007年，Hays和Efros发现，照片抠图后的背景填充可以从大量照片中学到
- AI的知识获取瓶颈可望采用机器学习算法从大数据中得到解决

深度学习的兴起（2006至今）

- 2006年，大牛Hinton在Science发文，首次将深度学习纳入科学家视野
- 2011年，微软的邓力采用深度学习模型把语音识别的错误率降低30%
- 2012年，在ImageNet评测中，AlexNet比采用传统机器学习算法（SVM）的第二名错误率降低42%（15% vs. 26%）
- 2012年，谷歌自动驾驶汽车获得了美国首个自动驾驶车辆许可证
- 2014年，Goodfellow 等人发表 Generative Adversarial Networks

深度学习的兴起（2006至今）

- Deep learning refers to a class of machine learning techniques, where many layers of nonlinear information processing stages in hierarchical architectures are exploited for pattern classification and for unsupervised feature learning (Li Deng, Microsoft Research)
- Deep learning learns **multiple levels of representations that correspond to different levels of abstraction (really?)**
- 深度学习在图像识别、语音处理、机器翻译、信息检索等方面都取得了最好（**state-of-the-art**）结果

2016年的AI重磅突破

- **DeepMind AlphaGo** computer program played an historic five game match against Lee Sedol in March 2016. AlphaGo won the match, making it the first time a computer Go program had defeated a world class human player on even terms.
- Nvidia introduced Tesla P100 GPU and DGX-1 Deep Learning Supercomputer (\$129,000) in April 2016.

AI突破

- GNMT is startlingly accurate (washington post, Oct 3, 2016)
- 2017年5月，DeepMind的AlphaGo战胜世界围棋冠军柯洁
- 2017年12月，采用零知识进行强化学习的**AlphaGo Zero**训练了34个小时就胜过了AlphaGo.

Abridged history of AI

- 1943 McCulloch & Pitts: Boolean circuit model of brain
- 1950 Turing's "Computing Machinery and Intelligence"
- 1956 Dartmouth meeting: "Artificial Intelligence" adopted
- 1952—69 Look, Ma, no hands!
- 1950s Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
- 1965 Robinson's complete algorithm for logical reasoning
- 1966—73 AI discovers computational complexity
Neural network research almost disappears
- 1969—79 Early development of knowledge-based systems
- 1980-- AI becomes an industry
- 1986-- Neural networks return to popularity
- 1987-- AI becomes a science
- 1995-- The emergence of intelligent agents
- 2001-- The availability of very large data sets
- 2006-- The emergence of deep learning

State of the art

- [Proverb solves crossword puzzles](#) better than most humans (1999)
- [Carnegie Mellon Tartan Racing Wins \\$2 Million DARPA Urban Challenge](#). A self-driving SUV called Boss made history by driving swiftly and safely while sharing the road with human drivers and other robots. (Nov 2007)
- [Proved a mathematical conjecture](#) (Robbins conjecture) unsolved for decades in Jan. 2011 (by Dr. Christoph Koutschan and Dr. Manuel Kauers from the Research Institute for Symbolic Computation of the Johannes Kepler University Linz, Austria). Producing the proof required vast computer resources and was only possible after the formula had been prepared for computer-assisted calculation.

State of the art

- [Microsoft demoed instant English-Chinese spoken translation](#) that keeps voice and accent in Tianjin in October, 2012
- [Japanese artificial intelligence](#) passes university exams (but not the notoriously selective University of Tokyo). The Todai Robot Project, and aims to pass the entrance exam for the University of Tokyo in 2021. ([Tokyo test](#)) (Nov 2015)
- [How Microsoft beat Google at understanding images with machine learning](#) (both beats human) in December 2015 (an [ImageNet test](#))
- **2018.3.15** 微软中-英机器翻译水平可“与人类媲美”
- 2018年4月，腾讯首次在博鳌论坛中进行AI同传实验

8:30

0.00K/s 99%



推文



Sebastian Ruder

@seb_ruder

Microsoft reports that they've achieved human parity on Chinese-to-English translation (27.40 BLEU; 1 BLEU better than best result of WMT 2017). Model is a Transformer (NIPS 2017) + Dual Learning (NIPS 2016) + Deliberation Nets (NIPS 2017). microsoft.com/en-us/research...

由  Microsoft 翻译自英文

微软报道说, 他们已经实现了汉英翻译中的人的平等 (27.40 蓝, 1 蓝比最好的结果 WMT 2017)。模型是一个变压器 (2017) + 双学习 (奶头 2016) + 审议网 (奶头 2017)。 microsoft.com/en-us/

发布回复推文

也不要
被胜利
冲昏了
头脑!

AI的分支学科

- 按研究方法分
 - 符号智能
 - 搜索
 - 推理
 - 知识工程
 - 计算智能（软计算）
 - 自然计算
 - 神经计算，进化计算、免疫计算、粒群计算、混沌理论
 - 模糊数学和粗糙集
 - 机器学习和数据挖掘
 - 模式识别
 - 智能Agent

AI的分支学科

- 按应用领域
 - 自然语言处理
 - 计算机视觉（图像/视频识别与理解）
 - 计算机听觉（语音识别/合成）
 - 机器人
 - 智能XX（控制、管理、决策...）

人工智能的思考

- 有人把人工智能的发展方向分为
 1. 运算智能（快速计算和记忆存储）
 2. 感知智能（视觉、听觉、触觉等感知能力）
 3. 认知智能（能理解会思考）这种说法合适吗？
- 李航说
 - 从情感、创造力和自由意志几个方面看，在现代计算机上实现强人工智能是非常困难的。

人工智能的思考

- 那么智能的本质是什么？
 - 苏格拉底：I know that I am intelligent, because I know that I know nothing.
 - 爱因斯坦：The true sign of intelligence is not knowledge but imagination.
 - 很多不同的智能类型：analytic, linguistic, emotional, creative, practical, learning, self-awareness...
 - 智能是涉身(embodiment)的吗？

AI领域著名会议（周志华）

- tier-1:

IJCAI (1+): International Joint Conference on Artificial Intelligence

- AAAI (1): National Conference on Artificial Intelligence

COLT (1): Annual Conference on Computational Learning Theory

CVPR (1): IEEE International Conference on Computer Vision and Pattern Recognition

ICCV (1): IEEE International Conference on Computer Vision

ICML (1): International Conference on Machine Learning

NIPS (1): Annual Conference on Neural Information Processing Systems

ACL (1-): Annual Meeting of the Association for Computational Linguistics

KR (1-): International Conference on Principles of Knowledge Representation and Reasoning

SIGIR (1-): Annual International ACM SIGIR Conference on Research and Development in Information Retrieval

SIGKDD (1-): ACM SIGKDD International Conference on Knowledge Discovery and Data Mining

UAI (1-): International Conference on Uncertainty in Artificial Intelligence

AI领域著名会议（周志华）

- **Tier-2:**

AAMAS (2+): International Joint Conference on Autonomous Agents and Multiagent Systems

ECCV (2+): European Conference on Computer Vision

ECML (2+): European Conference on Machine Learning

ICDM (2+): IEEE International Conference on Data Mining

SDM (2+): SIAM International Conference on Data Mining

ICAPS (2): International Conference on Automated Planning and Scheduling

ICCBR (2): International Conference on Case-Based Reasoning

COLLING (2): International Conference on Computational Linguistics

ECAI (2): European Conference on Artificial Intelligence

ALT (2-): International Conference on Algorithmic Learning Theory

EMNLP (2-): Conference on Empirical Methods in Natural Language Processing

ILP (2-): International Conference on Inductive Logic Programming

PKDD (2-): European Conference on Principles and Practice of Knowledge Discovery in Databases

AI领域著名会议（周志华）

- **Tier-3:**

ACCV (3+): Asian Conference on Computer Vision

DS (3+): International Conference on Discovery Science

ECIR (3+): European Conference on IR Research

ICTAI (3+): IEEE International Conference on Tools with Artificial Intelligence

PAKDD (3+): Pacific-Asia Conference on Knowledge Discovery and Data Mining

ICANN (3+): International Conference on Artificial Neural Networks

AJCAI (3): Australian Joint Conference on Artificial Intelligence

CAI (3): Canadian Conference on Artificial Intelligence

CEC (3): IEEE Congress on Evolutionary Computation

FUZZ-IEEE (3): IEEE International Conference on Fuzzy Systems

GECCO (3): Genetic and Evolutionary Computation Conference

ICASSP (3): International Conference on Acoustics, Speech, and Signal Processing

ICIP (3): International Conference on Image Processing

ICPR (3): International Conference on Pattern Recognition

IEA/AIE (3): International Conference on Industrial and Engineering Applications of Artificial Intelligence and Expert Systems

IJCNN (3): International Joint Conference on Neural Networks

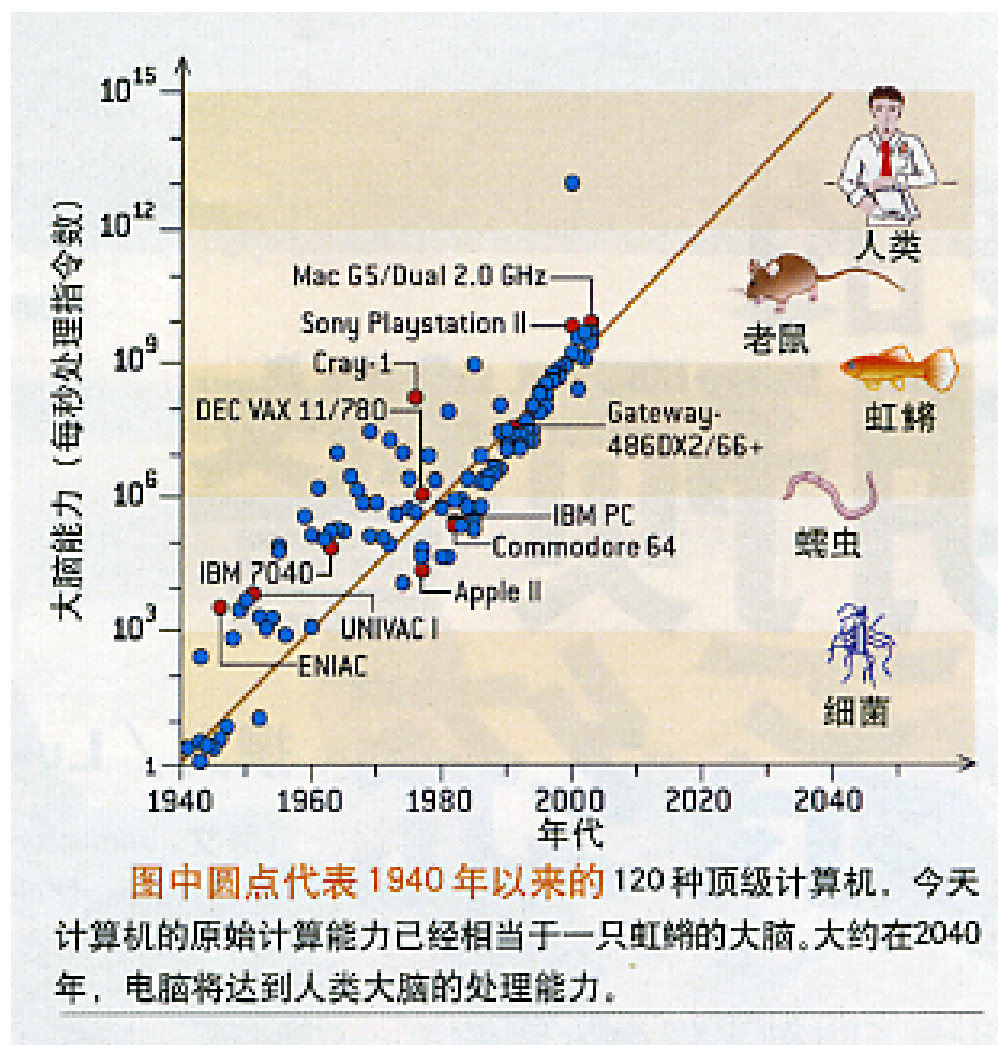
IJNLP (3): International Joint Conference on Natural Language Processing

PRICAI (3): Pacific-Rim International Conference on Artificial Intelligence

AI领域著名期刊

- Journal of Machine Learning Research
- Artificial Intelligence
- Artificial Intelligence Review
- AI Magazine
- Machine Learning
- Computational Intelligence
- Journal of AI Research
- Pattern Recognition
- Artificial Intelligence in Medicine
- IEEE Trans. on Pattern Analysis & Machine Intelligence
- IEEE Trans. on Knowledge & Data Engineering
- IEEE Trans. on Neural Networks
- Journal of Data Mining & Knowledge Discovery

计算机的发展



图来自史忠植教授的讲义

AI能造出超人？

- 最近AI的发展得益于计算能力的提高和大数据的易得
- Computers will be like humans by 2029:
Google's Ray Kurzweil <http://singularity.com/>
- Experts believe that AI will outperform humans in many activities in the near future, such as translating languages (by 2024), writing a high-school essay (by 2026), driving a truck (by 2027), working in retail (by 2031), writing a bestselling book (by 2049), and working as a surgeon (by 2053). (<https://arxiv.org/abs/1705.08807>)



AI Era is here

“Now this is not the end. It is not even the beginning of the end. But it is, perhaps, the end of the beginning.”

--Winston Churchill

Misuse of AI

- Lethal autonomous weapons
 - Cyber attack by AI systems
 - Adaptive blackmailing
 - Fake news/tweets by AI bots
 - Deepfake
-
- 360/各类管家/各种浏览器的风险！
 - 支付宝/微信支付的危险！
 - 隐私保护/数据完整性越来越重要

区块链？

人工智能与中国

- 2015年07月，国务院发布“互联网+”行动的指导意见，部署11大行动，其中就包括“互联网+”人工智能
- 2017年7月，国务院发布《新一代人工智能发展规划》
- 2017年10月，习近平总书记在十九大报告中强调：“推动人工智能与实体经济深度融合。”
- 2018年4月，教育部发布《高等学校人工智能创新行动计划》：到2020年建立50家人工智能学院、研究院或交叉研究中心，加大人工智能领域人才培养力度

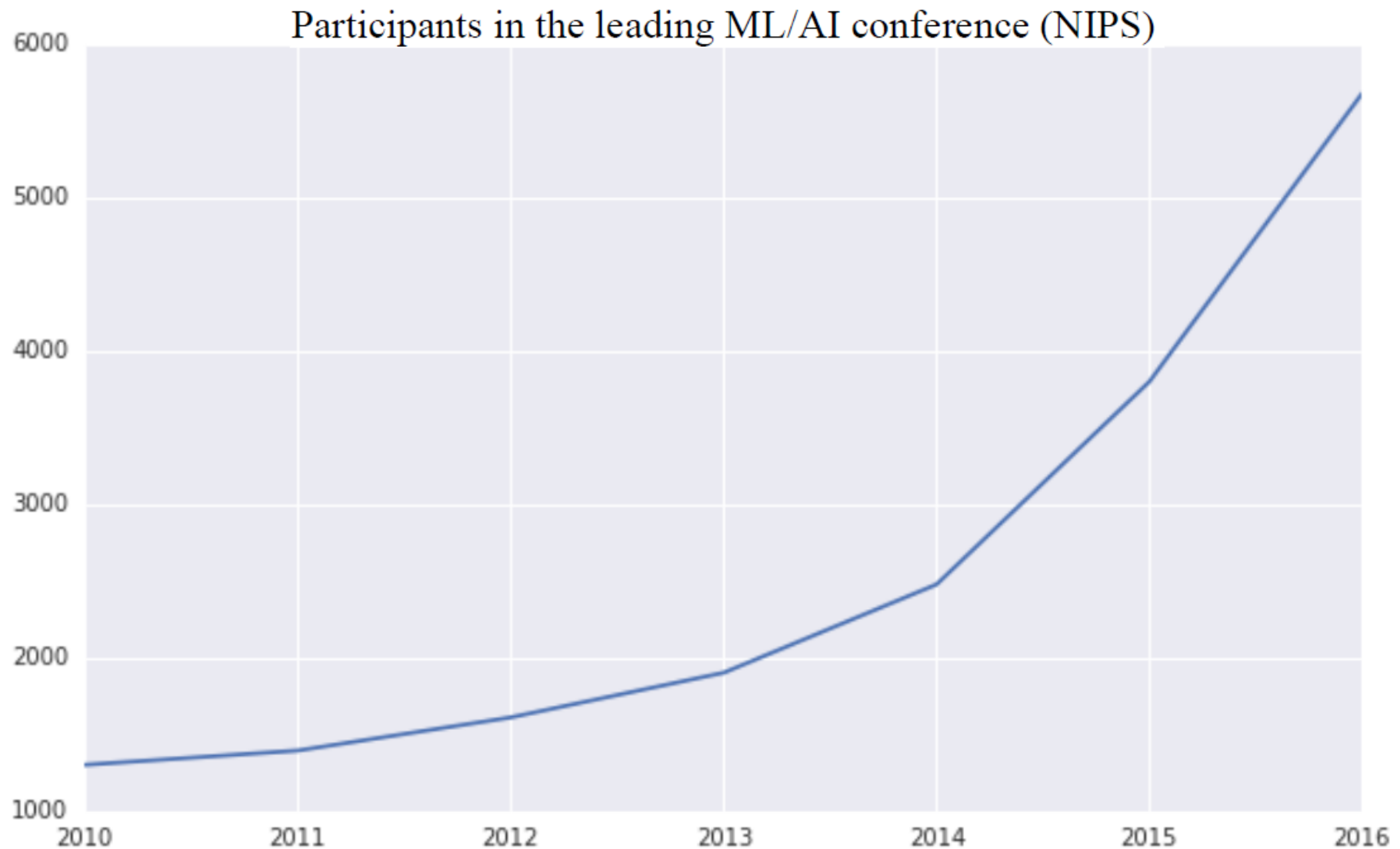
人工智能与中国

- 中国人工智能人才缺口超**500万** 供求比例仅为**1:10**（人民日报：<http://finance.people.com.cn/n1/2017/0714/c1004-29404223.html>）中国人工智能人才总量为世界第**7**，仅占美国**6%**，还不如印度
- **2018.8.7**，**26**高校（包括清华、南大、复旦、上交大、西交大、浙大、哈工大、东南大、武汉大、厦大等）首次在北航集会，提出申办人工智能本科专业。中央教育电视台、北京电视台、中国日报网、人民日报海外版等**10**多家媒体对此进行报道。

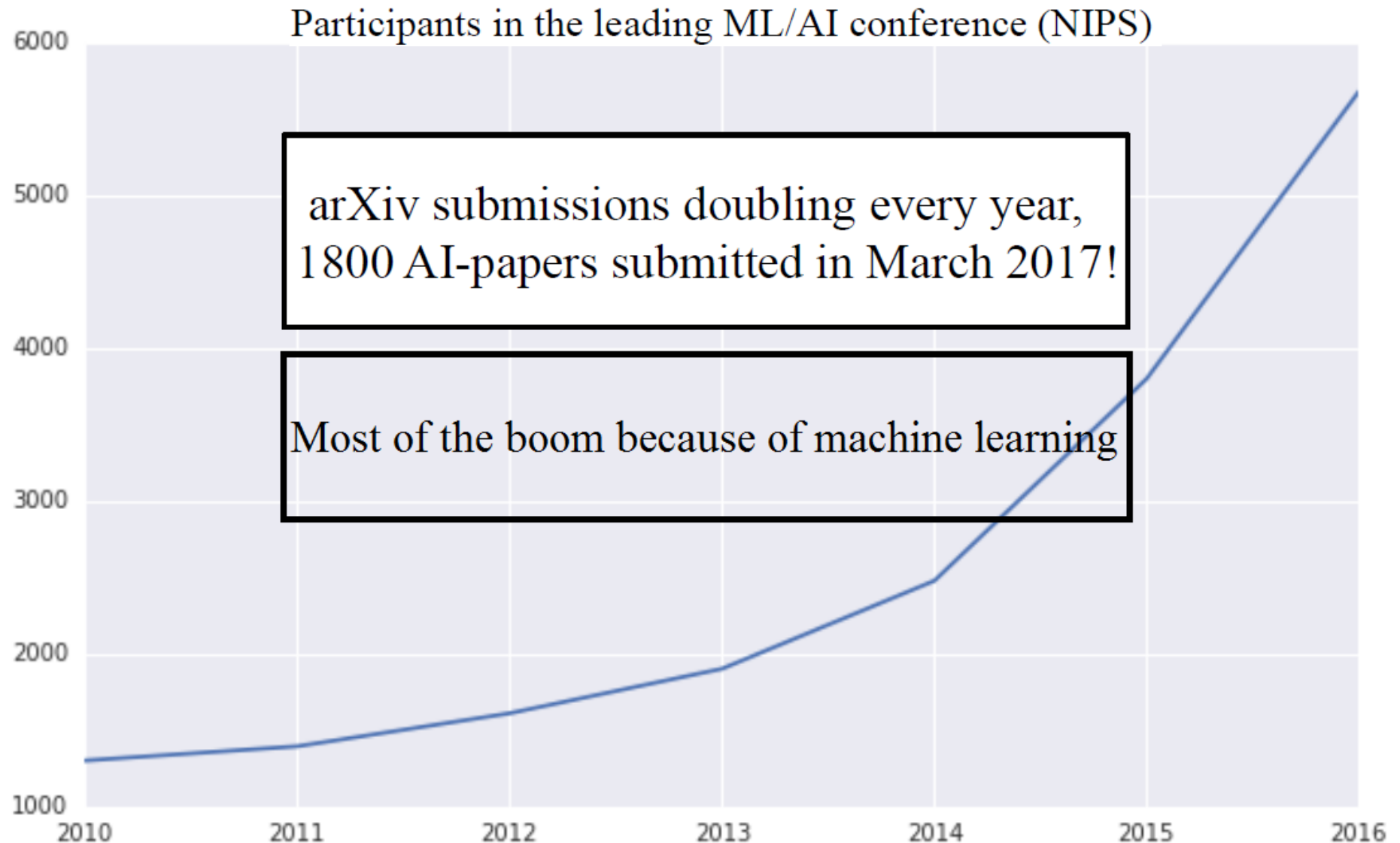
Course overview

- 授课计划:
 - 人工智能介绍
 - 智能体
 - 搜索
 - 不确定性和概率推理
 - 机器学习
 - 自然语言处理
 - 多模态学习

AI BOOM: THE ACADEMIC PERSPECTIVE



AI BOOM: THE ACADEMIC PERSPECTIVE



补充内容1

- 论文选读
 - 增加论文选读，加深对重要论文和目前国际研究现状的了解
 - 论文列表会出现在ppt中，大家可选择阅读，有时会就论文在课上提问，从而检查阅读情况

补充内容2

- 根据我的研究领域，适当补充一、两节课加以介绍（本次拟请外校老师讲）
- 前面讲过的内容有：
 - Web挖掘（数据挖掘分支）：目前WEB是许多AI应用的数据源（博客、微博等），如何从中挖掘数据，表示数据，值得研究
 - 简繁汉字转换：如何采用语言处理技术减少两岸交流障碍，助力和平发展
 - 基于云计算的协同式多语言翻译平台：利用自然语言处理技术改造传统的翻译产业
 - 少数民族语言处理：通过语言技术减少数字鸿沟
 - 神经机器翻译

课程设计

- 6个人左右一组，做1个课程设计。课程设计不限于课上讲述的内容，与书上的有关章节相关或与你们的研究方向相关的都可以。课程设计完成后，选一名代表做一个演讲。
- 课程设计必须给出实验报告或论文。实验报告要描述清楚细节；如果是论文，那么必须要基本达到能发表的要求。

成绩计算

- 成绩将参照平时作业（20）、期末考试（65）、课程设计（15分，其中演讲印象得1或2分）综合给出。
- 作业通过ftp上传到服务器，命名
姓名_学号_第N次作业.doc
其中N=1, 2, 3...

如何学好本课程

- 看书、做作业、读论文；打好基础，了解热点；根据自己的兴趣，选择合适的领域进行深入学习
- 跟随巨人的脚步
 - Google “ten most important people in AI”
 - ...
- 站上巨人的肩膀，更上一层楼！
 - 让厦大以你为荣！

作业

- 阅读图灵关于AI的1950年的论文。请你结合AI几十年来的发展，对他的主要观点进行评述。特别是通过图灵测试的可能性。
- 阅读文章
 - **Stuart Russell**, on the Future of Artificial Intelligence 2003, <https://ubiquity.acm.org/article.cfm?id=964695>
 - Executive Office of the President, National Science and Technology Council, PREPARING FOR THE FUTURE OF ARTIFICIAL INTELLIGENCE, 2016
 - 结合这两年人工智能的热门，谈谈你对人工智能看法

作业

- 查询AI文献，看看如下的任务现在计算机能否解决，并找出其难点且对其解决的年代进行预测：
 - 乒乓球比赛
 - 在市中心开车
 - 去超市买东西
 - 在网上买东西
 - 参加桥牌比赛
 - 发现并证明数学定理
 - 写一个有趣的故事
 - 从英语到西班牙语的口语实时翻译
 - 完成复杂的外科手术

作业

- 阅读相关论文，看看在AlphaGo中，采取了那些新的人工智能技术？特别与1997年深蓝战胜Kasparov的AI技术做比较。

论文选读

- **A. Turing, 1950**, Computing Machinery and Intelligence
- **N. Nilson, 1991**, Logic and artificial intelligence. In *Artificial Intelligence* 47 (1991) 31-56
- **Thomason, Richmond**, "Logic and Artificial Intelligence", *The Stanford Encyclopedia of Philosophy* (Spring 2009 Edition)
<http://plato.stanford.edu/entries/logic-ai/>

论文选读

- **G. E. Hinton, 2007**, "Learning multiple layers of representation," Trends in Cognitive Sciences, 11, pp. 428–434.
- **David Silver et al, 2016**, Mastering the game of Go with deep neural networks and tree search, Nature 529, 484–489 (28 January 2016)